TOPIC 02

Probability Basics

The language of chance: events, rules, conditional probability, Bayes and random variables

WHAT YOU WILL LEARN

- Define experiments, sample spaces and events
- Use the complement, addition and multiplication rules
- Work with conditional probability, independence and Bayes' theorem
- Count outcomes with permutations and combinations
- Describe a random variable by its expected value and variance

1. Experiments, outcomes and events

Probability measures how likely something is to happen when the result is uncertain. We start with a few precise words.

Term	Meaning	Rolling one die
Random experiment	A process whose result is uncertain but whose possible results are known	Roll the die once
Outcome	One possible result	3
Sample space, S	The set of all possible outcomes	S = {1, 2, 3, 4, 5, 6}
Event	Any collection of outcomes (a subset of S)	'Even number' = {2, 4, 6}
Simple event	An event with exactly one outcome	{5}

Events can be combined. Write **A ∪ B** for 'A or B (or both)', **A ∩ B** for 'A and B', and $\mathbf{A^c}$ for 'not A' (the complement).

2. What does probability mean?

Approach	Idea	Formula / example
Classical	All outcomes equally likely, so count	P(A) = outcomes in A / outcomes in S. P(even) = 3/6 = 0.5
Empirical (frequency)	Repeat many times and see the long-run proportion	P(A) ≈ times A happened / trials
Subjective	A degree of belief based on evidence	'70% chance of rain tomorrow'

The empirical view is justified by the **Law of Large Numbers**: as the number of trials grows, the observed proportion settles at the true probability.

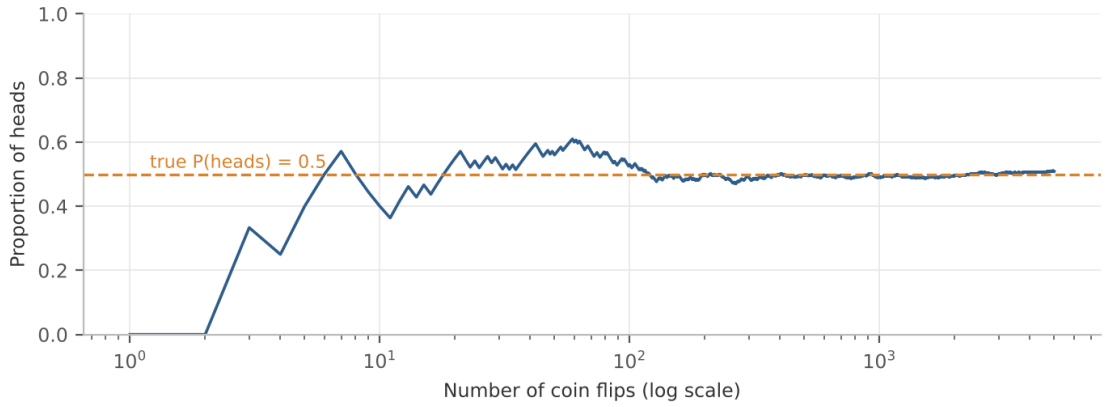


Figure 1. The running proportion of heads wobbles wildly after a few flips, then settles near 0.5.

COMMON MISTAKE **The gambler's fallacy**

After five heads in a row the next flip is **still** 50/50. The Law of Large Numbers says proportions settle over the long run; it does not say the coin 'owes' you tails.

3. The rules of probability

DEFINITION **Axioms**

- $0 \le P(A) \le 1$ for every event A.
- $P(S) = 1$. Something in the sample space must happen.
- If A and B cannot both happen (mutually exclusive), $P(A \cup B) = P(A) + P(B)$.

Complement rule

$$P(A^C) = 1 - P(A)$$

Useful whenever 'at least one' is easier as 1 − 'none'. Example: the chance of at least one six in four rolls of a die is $1 - (5/6)^4 = 0.5177$.

Addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

We subtract $P(A \cap B)$ because outcomes in both events were counted twice. If A and B are mutually exclusive the last term is 0.

WORKED EXAMPLE **Cards**

Draw one card from a standard 52-card deck. P(heart) = 13/52, P(king) = 4/52, P(heart and king) = 1/52 (the king of hearts). So P(heart or king) = 13/52 + 4/52 − 1/52 = 16/52 = 0.3077.

4. Conditional probability and independence

DEFINITION **Conditional probability**

The probability of A **given that** B has happened. We shrink the sample space to B and ask how much of B is also A.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0$$

Multiplication rule

$$P(A \cap B) = P(B)\,P(A \mid B)$$

WORKED EXAMPLE **Two aces without replacement**

Draw two cards without putting the first back. P(first ace) = 4/52. Given that, only 3 aces are left in 51 cards, so P(second ace | first ace) = 3/51. Therefore P(both aces) = (4/52)(3/51) = 0.00452 (about 1 in 221).

Independence

Events A and B are **independent** if knowing one tells you nothing about the other: P(A | B) = P(A), which gives the simple product rule **P(A ∩ B) = P(A) P(B)**. Rolling a die twice gives independent results; drawing cards without replacement does not.

COMMON MISTAKE **Independent is not the same as mutually exclusive**

Mutually exclusive events cannot happen together, so knowing one happened tells you the other did **not**. That is the opposite of independence. Two events with positive probability that are mutually exclusive are always dependent.

5. Bayes' theorem

Conditional probability can be turned around. If we know P(B | A) and want P(A | B), Bayes' theorem does it:

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)} = \frac{P(B \mid A)\,P(A)}{P(B \mid A)P(A) + P(B \mid A^c)P(A^c)}$$

WORKED EXAMPLE **A medical test**

A disease affects 1% of people. A test is positive for 90% of people who have it (sensitivity) and also for 5% of people who do not (false-positive rate). You test positive. What is the chance you have the disease?

- P(positive) = 0.90 × 0.01 + 0.05 × 0.99 = 0.0585
- P(disease | positive) = 0.009 / 0.0585 = **0.154**, about 15%

Surprisingly low, because healthy people vastly outnumber sick ones, so most positive results are false alarms. A table of 10,000 people makes this concrete:

	Positive	Negative	Total
Has disease	90	10	100
No disease	495	9,405	9,900
Total	585	9,415	10,000

Of 585 positives only 90 are truly ill: 90/585 = 0.154.

6. Counting: permutations and combinations

When outcomes are equally likely, probability reduces to counting. These two tools are enough for the binomial distribution in Topic 03.

	Order matters?	Formula	Example
Permutations	Yes	${}_nP_r = n! / (n - r)!$	Gold, silver, bronze among 8 runners: 336
Combinations	No	${}_nC_r = n! / (r!(n - r)!)$	Choose 3 of 8 for a committee: 56

Example: the chance of getting exactly 2 heads in 4 fair flips is $C(4,2) / 2^4 = 6/16 = 0.375$.

7. Random variables

> **DEFINITION** **Random variable**
>
> A rule that assigns a number to each outcome of a random experiment. Written with a capital letter (X), with specific values in lowercase (x). **Discrete**: takes countable values (number of heads). **Continuous**: takes any value in an interval (waiting time).

Expected value and variance

For a discrete random variable with probabilities $P(X = x)$:

$$E(X) = \mu = \sum x\,P(x), \qquad \mathrm{Var}(X) = \sigma^2 = \sum (x-\mu)^2 P(x) = E(X^2) - \mu^2$$

> **WORKED EXAMPLE** **One fair die**
>
> $E(X) = (1+2+3+4+5+6)/6 =$ **3.5**. $\mathrm{Var}(X) = ((-2.5)^2 + (-1.5)^2 + (-0.5)^2 + 0.5^2 + 1.5^2 + 2.5^2)/6 = 17.5/6 =$ **2.9167**, so $\sigma = 1.708$.

For the **sum of two dice** (Topic 03) the same calculation gives $E = 7$ and $\mathrm{Var} = 5.8333 = 35/6$. Notice that both are exactly double the one-die values, an example of the rules below.

Rule	Statement
Linear change	$E(aX + b) = aE(X) + b$ and $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$
Sum	$E(X + Y) = E(X) + E(Y)$ always
Variance of a sum	$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ **if X and Y are independent**

The variance of a sum rule is the seed of the Central Limit Theorem and of the standard error formula $\sigma/\sqrt{n}$.

Law of Large Numbers, restated

The sample mean $\bar{x}$ of independent observations gets closer to $\mu = E(X)$ as n grows. This is why a large sample is a better estimate.

8. In Python

```
import itertools, math
from fractions import Fraction

S = list(itertools.product(range(1, 7), repeat=2))           # sample space of two dice
A = [o for o in S if sum(o) == 7]                             # event: sum is 7
print(Fraction(len(A), len(S)))                               # 1/6

print(math.comb(4, 2) / 2**4)                                 # 0.375
p_pos = 0.9*0.01 + 0.05*0.99
print(0.9*0.01 / p_pos)                                       # Bayes: 0.1538
```

SUMMARY

- Probability lives between 0 and 1; the sample space sums to 1.
- Complement: $P(A^c) = 1 - P(A)$. Addition: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Conditional: $P(A \mid B) = P(A \cap B) / P(B)$. Independent events satisfy $P(A \cap B) = P(A)\,P(B)$.
- Bayes reverses a conditional and often defeats intuition when the base rate is low.
- A random variable has a mean $E(X)$ and a variance $Var(X)$, and variances of independent variables add.

Practice questions

Try each question before reading the answer underneath it.

Q1. A bag has 5 red and 3 blue balls. Two are drawn without replacement. Probability both are red?

Answer. $(5/8)(4/7) = 20/56 = 0.3571$.

Q2. $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.2$. Are A and B independent?

Answer. $P(A)P(B) = 0.2 = P(A \cap B)$, so yes.

Q3. $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.2$. Find $P(A \cup B)$ and $P(A \mid B)$.

Answer. $P(A \cup B) = 0.4 + 0.5 - 0.2 = 0.7$. $P(A \mid B) = 0.2/0.5 = 0.4$.

Q4. Probability of at least one head in 3 fair coin flips?

Answer. $1 - (1/2)^3 = 7/8 = 0.875$.

Q5. A game pays 10 with probability 0.1, 2 with probability 0.4 and 0 otherwise. Expected payout?

Answer. $E = 10(0.1) + 2(0.4) + 0(0.5) = 1.8$.

Q6. If $Var(X) = 9$, what is $Var(2X + 5)$ and the new standard deviation?

Answer. $Var = 2^2 \times 9 = 36$; $SD = 6$. Adding 5 does not change the spread.